

Voltage Sag Source Location Methods Performance Under Symmetrical and Asymmetrical Fault Conditions

Roberto Chouhy Leborgne, *Member, IEEE*, Daniel Karlsson, *Senior Member, IEEE*, and Jaap Daalder

Abstract—When several utilities operate interconnected power systems, a voltage sag generated in the grid of one utility may affect the loads connected to other utilities. From this point of view, it has become a crucial issue to identify the location of the source of a voltage sag. This paper describes and compares the methods for voltage sag source location. The methods are applied to locate the source of voltage sags considering symmetrical and asymmetrical faults. Radial and meshed areas are also analyzed. The results show that the available methods are not totally reliable and more research is needed to extend the location methods to meshed systems.

Index Terms-- Power quality, voltage sags (dips), source location.

I. NOMENCLATURE

DP:	disturbance power
DE:	disturbance energy
DPE:	disturbance power and energy
SST:	slope of system trajectory
RCC:	real current component
RS:	resistance sign
DR:	distance relay algorithm

II. INTRODUCTION

A voltage sag is a reduction of the rms voltage to a value below 90 % of the nominal voltage for a short period. The sag duration is related to the fault-clearing time, therefore, it does not exceed 1 minute [1].

Voltage sags are awakening the attention of utilities' engineers and end-users due to their impact on sensitive industrial loads; causing damage which can be estimated in billions of dollars per year [2]. The most relevant voltage sags are caused by faults in the power system. Unfortunately the sags are not confined to the fault location; they propagate through the system affecting loads connected far away from the fault location.

Considering this background the accountability for the generation of disturbances on the system must be assessed, and

it is necessary to classify the disturbance source as internal or external for a given system or utility.

Fault location is not a new issue on power systems research [3]. However, the perspective of sag source location motivated new investigations. The main difference with the earlier works is that now it is not wanted to know the exact location of the fault for protection purposes but the relative location of the fault for a given monitor position.

One of the first works [4] aiming to locate the source of the disturbance defined the concepts of "disturbance power and disturbance energy". These concepts were applied to locate the origin of disturbances caused by capacitor switching and faults.

Later, a second work introduced the concept of "slope of the system trajectory" to locate the origin of voltage sags [5]. This method is based on the analysis of the current and the voltage component in the current direction. Later, the method was generalized to a new approach using the sign of an estimated resistance to classify the source of voltage sags [6].

Meanwhile, another approach using the variation of the "real current component" was introduced [7]. This method uses the current component on the voltage direction to classify the sag source as upstream or downstream to a given monitor location.

The increasing use of digital relays motivated an extended applicability of these devices for power quality analysis. The distance relay algorithms can be used for the detection of the voltage sag source, based on the magnitude and angle of the measured impedance [8].

All the previous methods need both the current and voltage measurements to locate the source of the sag. Alternatively a method based only on the voltage magnitude and phase-angle jump was proposed for the classification of the source in the connection point of end-users. In this case, the location of the sag source is classified as internal or external to the end-user grid [9].

The proposed methods for sag source location have so far limited validity as conditions as asymmetrical faults and meshed grids have not been extensively analyzed. This paper describes a comprehensive investigation of the performance of the methods for the location of the sag source.

III. VOLTAGE SAG SOURCE LOCATION METHODS

The location of the sag source as observed from a monitored bus is defined as upstream and downstream, using the pre-fault power flow direction as indicator. Upstream is the region against the flow of power and downstream is the region

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following the flow of power, as shown in Fig. 1.

A. Disturbance power and energy

The first published method aiming to locate the source of the disturbances is based on the disturbance power and the disturbance energy [4]. The disturbance power is defined as the difference between the power delivered during the event and the steady-state delivered power. The disturbance energy results from the time integral of the disturbance power.

$$DP = P_f - P_{ss} \quad (1)$$

$$DE(t) = \int_0^t DP(u) du \quad (2)$$

where P_f and P_{ss} are the power delivered during the fault and during the steady-state condition.

This method prescribes that the final sign of the disturbance energy defines the location of the sag source. If the final value of the disturbance energy is greater than 80% of the peak value this test is conclusive. Then, if the final disturbance energy is positive the sag source is located downstream. If the final disturbance energy is negative the source of the sag is located upstream. If the sign of the initial peak of the disturbance power coincides with the sign of the final disturbance energy a high degree of confidence is expected [4].

B. Slope of system trajectory

The method of the “slope of the system trajectory” [5] is based on the relation between the voltage and the current shown in Fig. 1. For instance, for a fault located downstream of the monitored point, the active power measured at the PQ-monitor is

$$VI \cos \theta = -R_1 I^2 + E_1 I \cos \theta_1 \quad (3)$$

where E_1 is the voltage at the source 1, R_1 is the real part of Z_1 , V and I are the rms voltage and current measured by the PQ-monitor, θ is the angle between voltage and current at the monitored location and θ_1 is the angle at the source E_1 .

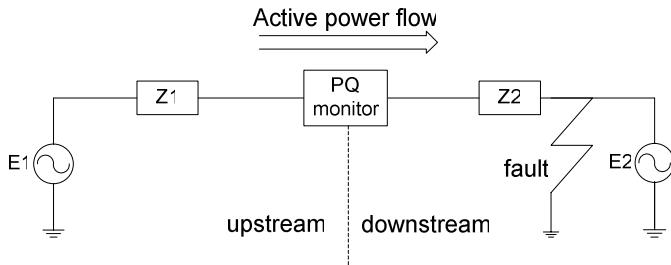


Fig. 1. Basic circuit to describe the localization of the sag source

Considering that $V \cos(\theta) > 0$, this expression can be substituted by $|V \cos(\theta)|$. Both parts of (3) can be divided by I to obtain (4). This relation is a line with slope $-R$.

$$|V \cos \theta| = -R_1 I + E_1 \cos \theta_1 \quad (4)$$

On the other hand, the authors claim that if the fault is upstream, a similar linear equation is obtained but the slope is $+R$, as shown in (5).

$$|V \cos \theta| = +R_1 I + E_1 \cos \theta_1 \quad (5)$$

Applying least-square method on a set of values of I and $|V \cos(\theta)|$ during a voltage sag, the points are linear fitting. The sign of the slope of this line indicates the position of the fault: positive slope indicates upstream fault and negative slope means downstream fault. This method also considers a possible inversion of the power flow during the sag. An inversion of the power flow indicates an upstream fault. This method is clear for symmetrical faults, here positive sequence voltage and positive sequence current are used. The method has not been extensively tested for asymmetrical faults. It is possible that the slopes have different sign for each phase and this method may not be conclusive.

C. Resistance sign

An extended method later developed [6] proposes to use the sign of a calculated resistance to classify the source location of the sag. This method uses the positive-sequence voltage and current. The resistance is estimated using (6) and (7), taking a set of n measured currents and voltages including pre-event and during-sag values. If the two estimated resistances using (6) and (7) have the same sign the test is conclusive. A positive sign means an upstream fault whereas a negative sign indicates a downstream fault.

$$\begin{bmatrix} R \\ X \\ E_{RE} \end{bmatrix} = \begin{bmatrix} I_{RE}(1) & I_{IM}(1) & 1 \\ \vdots & \vdots & \vdots \\ I_{RE}(n) & I_{IM}(n) & 1 \end{bmatrix}^{-1} \begin{bmatrix} V_{RE}(1) \\ \vdots \\ V_{RE}(n) \end{bmatrix} \quad (6)$$

$$\begin{bmatrix} R \\ X \\ E_{IM} \end{bmatrix} = \begin{bmatrix} I_{IM}(1) & I_{RE}(1) & 1 \\ \vdots & \vdots & \vdots \\ I_{IM}(n) & I_{RE}(n) & 1 \end{bmatrix}^{-1} \begin{bmatrix} V_{IM}(1) \\ \vdots \\ V_{IM}(n) \end{bmatrix} \quad (7)$$

where the subscripts RE and IM denote the real and imaginary part. The matrix power (-1) is the pseudo-inverse of the matrix.

In order to apply this method, the power flow must be observed. If the power flow is reversed during the sag, the voltages and currents used in (6) and (7) must only include values before the reversion of the power flow.

D. Real current component

The method of real current component [7] is based on the analysis of the variation of the real part of the current ($I \cos(\theta)$) having the voltage as reference. This approach starts also with equation (4). Here the authors claim that if at the beginning of the fault $I \cos(\theta) > 0$ then the fault is located downstream. On the other hand if at the beginning of the fault $I \cos(\theta) < 0$ then the fault is upstream. Basically, what this method proposes is to investigate the power factor ($\cos(\theta)$) at the beginning of the sag. If $-\pi/2 < \theta < \pi/2$ then the fault is considered to be downstream, otherwise it is considered upstream. The analysis is done using phase values and it is expected to obtain the same indication in each of the three phases, even for asymmetrical faults.

E. Distance relay algorithm

The approach of using a distance relay algorithm [8] is slightly different from the former methods. The basic idea is to analyze the impedance seen during the event. The impedance is estimated using the voltage and current phasors at the monitored location. For a downstream fault the impedance value can be calculated by

$$Z_{seen} = \frac{\bar{V}}{\bar{I}} = Z_2 + \Delta Z \quad (8)$$

where $\bar{V}$ and $\bar{I}$ are the measured voltage and current phasors, ΔZ is a function of the fault resistance and the load characteristics. For an upstream fault the current direction is reversed and the impedance will change in magnitude and angle. Therefore, this method proposes the following rule for the sag source detection: if $|Z_{SAG}| < |Z_{PRE-SAG}|$ and $angle(Z_{SAG}) > 0$ then the sag source is located downstream, else the sag source is located upstream.

F. Voltage-based method

A method based on the analysis of the voltage is presented in [9]. The approach estimates the voltage sag magnitude and the phase-angle jump. It is concluded that for a typical industrial installation the sags generated in the utility grid and the sags generated inside the industrial grid will follow different phase-angle jump vs. magnitude relations, and that the difference is larger for deeper sags. This method is not intended to classify the sag source at the interconnecting point of two utility grids; therefore, it will not be investigated in this paper.

IV. SIMULATIONS

In order to test the performance of the methods to locate the sag source a set of simulations are implemented in a transmission network. The simulation is realized using the PSCAD/EMTDC program. The faults are located in 6 buses and also 6 monitoring points are selected, as shown in Fig. 4.

The monitors M2 and M3 are located at the boundary of two radial 138 kV sub-transmission grids. The monitors M1, M4, M5, and M6 are located in the boundaries of a meshed 138 kV sub-transmission network. The transformers primary sides are connected to a 230 kV grid. The 230 kV transmission and the 138 kV sub-transmission grids are owned by two different utilities, therefore it is relevant to locate the source of the sag taking into account the borders between the two utilities.

The faults are distributed in both the 138 kV and the 230 kV networks, in upstream and downstream locations referred to the monitored buses. For the meshed networks the concept of upstream/downstream location is not straightforward defined and a more general classification as inside/outside the system is convenient. Therefore the result for the monitor locations M1, M4, M5, and M6 will not be presented.

It is necessary to recall that upstream/downstream is related to the active power flow. In the monitored buses M2 and M3 the active power flow is from the 230 kV system to the 138 kV

system at these buses. Therefore, upstream is equivalent with outside and downstream is equivalent with inside when the 138 kV system is taken as the reference.

At each fault location three-phase-to-ground and single-line-to-ground faults are simulated. All the sag source location algorithms are implemented at each monitored bus. The results are discussed for symmetrical faults, asymmetrical faults, and meshed systems.

V. RESULTS

A. Symmetrical faults

First a fault located at F1 as shown in Fig. 4, is simulated. This fault is located upstream for both M2 and M3 monitored buses.

The results applying the method of disturbance power and energy (DPE) is presented in Fig. 2. Here the fault is classified upstream at both buses, as indicated by the negative disturbance energy at both monitored buses.

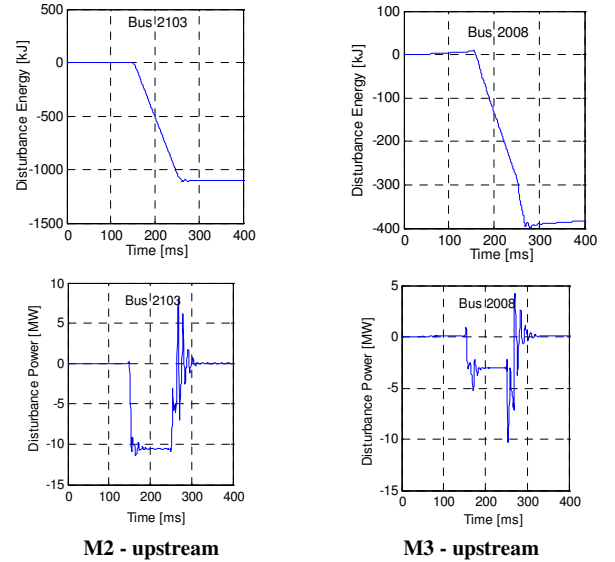


Fig. 2. Results from the method of disturbance power and energy, symmetrical fault at F1

The method of the slope of the system trajectory is applied and the result is shown in Fig. 3. Considering that a symmetrical fault is simulated, the method is applied for the positive-sequence of voltage and current. It can be seen in the figure that the slope of the system trajectory is positive at both monitored buses, meaning that the fault is located upstream.

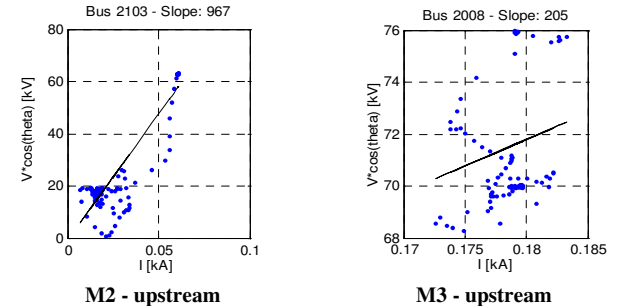


Fig. 3. Results from the method of slope of system trajectory, symmetrical fault at F1

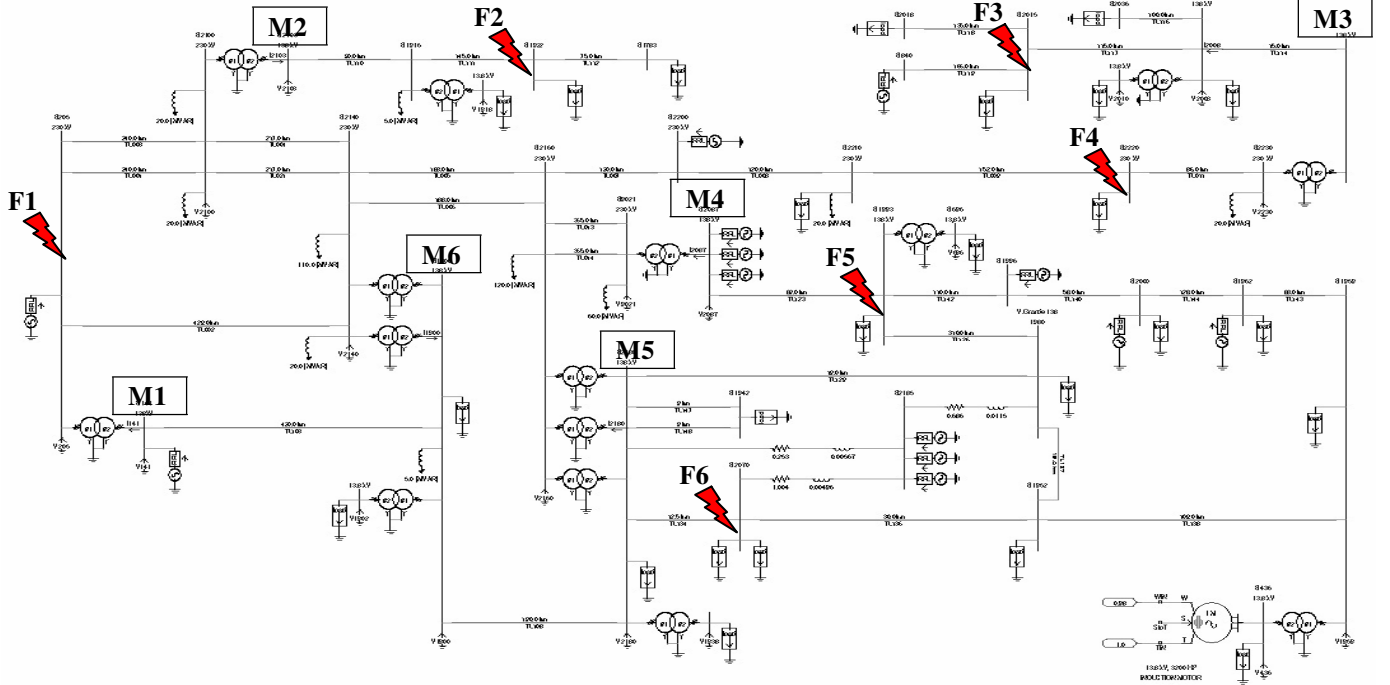


Fig. 4. Power system single-line diagram, fault locations and monitoring buses are indicated

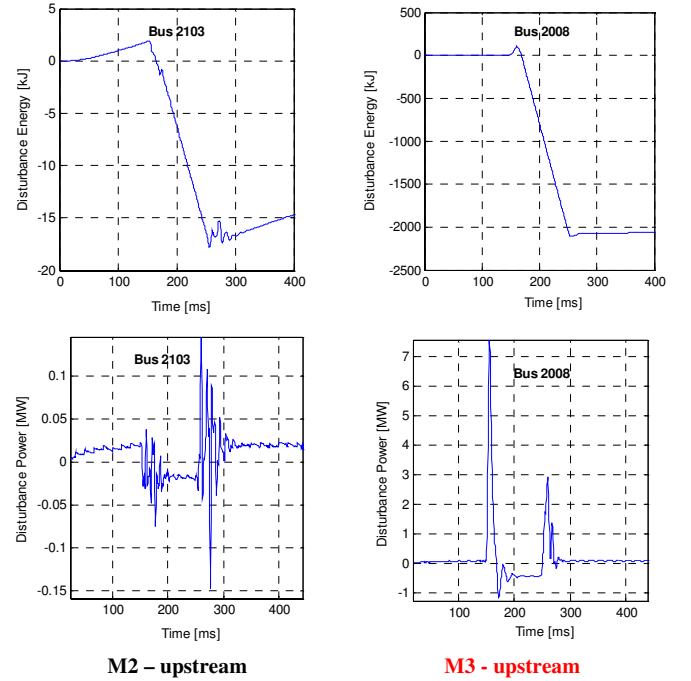
The results of the method of real current component, resistance sign and distance relay are shown in Table I. The method of the resistance sign applied at the monitored bus M2 classifies the source correctly, but when applied at the bus M3 the result is not conclusive because there is not sign agreement on the resistance calculated by using (6) and (7). The methods of real current component and distance relay locate correctly the sag source as upstream at the two buses.

TABLE I
OTHER RESULTS FOR SYMMETRICAL FAULT AT F1

	M2	M3
RCC	upstream	upstream
R (eq. 6) [ohm]	706	-524
R (eq. 7) [ohm]	313	60
RS	upstream	Not conclusive
$Z_{pre-sag}$ [ohm]	1320	445
Z_{sag} [ohm]	1320	415
$arg(Z_{sag})$ [deg]	-40	-21
DR	upstream	upstream

A second fault located at F3 is analyzed. Now the fault location is upstream with respect to M2 and downstream to M3. The location obtained using the method of disturbance power and energy is shown Fig. 5. The disturbance energy is negative for both buses indicating an upstream fault location. However, the disturbance power at M3 shows a positive peak reducing the confidence of the result for this bus. In fact the correct result is that the location is downstream at M3. So the method fails to correctly indicate the location of the faults, but the confidence of the result is low, indicating that the test was

not truly conclusive.



M2 – upstream

M3 - upstream

Fig. 5. Results from the method of disturbance power and energy, symmetrical fault at F3

The method of the slope of the system trajectory is applied to the positive sequence voltage and current. The result shows a positive slope at M2 and a negative slope at M3, indicating that the fault is located upstream at M2 and downstream at M3, as shown in Fig. 6. In this case, the method of the system trajectory locates correctly the fault with respect to the two monitored buses.

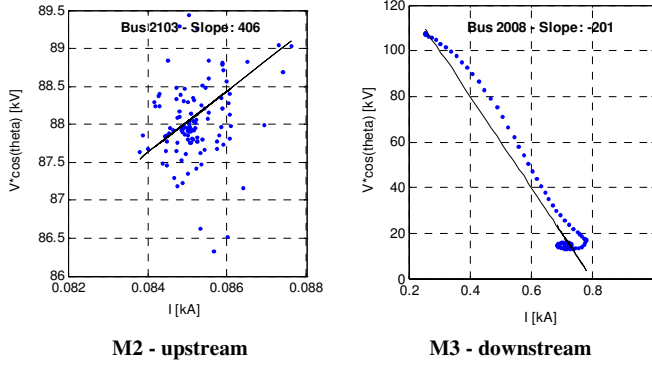


Fig. 6. Results from the method of slope of system trajectory, symmetrical fault at F3

Table II presents the results of the other methods. The method of resistance sign is not conclusive at M3 because two different signs are obtained. The methods of real current component and distance relay correctly classify the location of the fault at M2 and M3.

TABLE II
OTHER RESULTS FOR SYMMETRICAL FAULT AT F3

	M2	M3
RCC	upstream	downstream
R_I (eq. 6) [ohm]	866	41
R_I (eq. 7) [ohm]	273	-38
RS result	upstream	Not conclusive
$Z_{pre-sag}$ [ohm]	1324	450
Z_{sag} [ohm]	1324	60
$arg(Z_{sag})$ [deg]	-39	70
DR result	upstream	downstream

In order to test the methods more extensively four symmetrical faults are simulated at the locations F2, F4, F5, and F6, as shown in Fig. 4. The result of this analysis is presented in Table III. The second column shows the real location of the faults and the next columns show the results for each of the applied methods. It can be seen that most of the methods accurately classify symmetrical faults. However the method of resistance sign is not conclusive in 4 of the 8 cases and it gives the incorrect result in one of the 8 cases.

In general we can conclude that most of the methods give accurate results for symmetrical faults. Summarizing, the number of accurate results is 52 of 60 cases.

B. Asymmetrical faults

Here the methods are tested for a single-line-to-ground fault in phase A at the same 6 fault locations.

Asymmetrical faults are more difficult to locate because each phase shows a particular behavior. Fig. 7 shows the result of the method of the slope of the system trajectory for a fault located at F1. The results present different signs for different phases at M2. Therefore, no conclusive result is obtained at this bus. At M3 the estimated slope is positive for the three phases, so the fault is classified as upstream, agreeing with the real location.

TABLE III
SYMMETRICAL FAULTS AT OTHER LOCATIONS

Monitor M2						
	Location	DPE	SST	RCC	RS	DR
F2	DS	DS	DS	DS	NC	DS
F4	US	US	US	US	US	US
F5	US	US	US	US	US	US
F6	US	US	US	US	NC	US

Monitor M3						
	Location	DPE	SST	RCC	RS	DR
F2	US	US	US	US	DS	US
F4	US	US	US	US	US	US
F5	US	US	US	US	NC	US
F6	US	US	US	US	NC	US

US: Upstream, DS: Downstream, NC: No conclusive

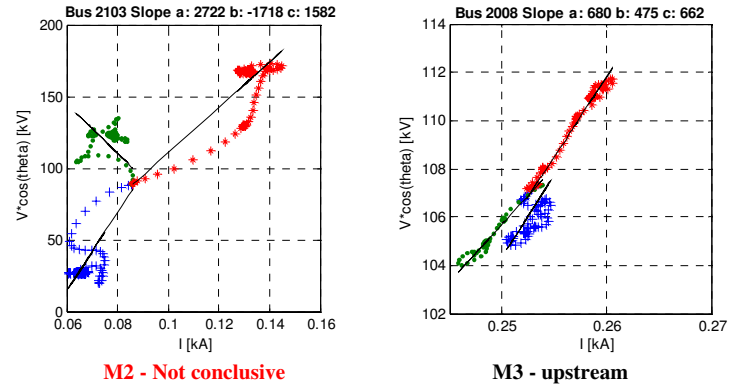


Fig. 7. Method of slope of system trajectory, asymmetrical fault at F1

TABLE IV
ASYMMETRICAL FAULTS AT OTHER LOCATIONS

Monitor M2						
	Location	DPE	SST	RCC	RS	DR
F1	US	DS	NC	NC	US	US
F2	DS	DS	NC	DS	NC	DS
F3	US	US	US	US	NC	US
F4	US	US	US	US	US	US
F5	US	DS	US	NC	NC	US
F6	US	DS	US	NC	US	DS

Monitor M3						
	Location	DPE	SST	RCC	RS	DR
F1	US	US	US	NC	US	US
F2	US	US	NC	US	US	US
F3	DS	NC	NC	DS	NC	DS
F4	US	US	US	NC	NC	US
F5	US	US	US	NC	US	US
F6	US	US	US	NC	US	US

US: Upstream, DS: Downstream, NC: No conclusive

The results for the 6 fault locations are summarized in Table IV. The method of distance relay presented the best

performance, obtained the correct location for the sag source for 11 of the 12 cases. The other methods have obtained accurate results in 8 cases (DPE), 8 cases (SST), 7 cases (RS), and 5 cases (RCC).

C. Further discussion

The loads under the monitors M2 and M3 were modeled as constant impedance. This is quite an unrealistic situation. Therefore, the analysis must be extended to other load characteristics, such as induction machines. Moreover, due to the high increase of distributed generation would be interesting to analyze the performance of the methods considering the effect of connecting distributed generation under the monitor locations.

The method of resistance sign was claimed to be tested for active loads, and the authors recommended considering when there is an inversion of the active power flow during the sag in order to choose the right voltage and current values to apply the method [6]. However, our results show that this method is very sensitive to the voltages and currents values chosen and in many of the simulated cases it fails to find the sag source location.

The classification of the sag source in a meshed grid is not straightforward as it is for the radial parts of the system. The fault currents flow through the monitored points and one could try to implement the same methods. However, the loops formed by a meshed system make the definition of upstream/downstream ambiguous.

Now we are working to develop algorithms to classify the sag source location for sags that triggered monitors at the meshed part of the system: M1, M4, M5, and M6.

VI. CONCLUSIONS

Several methods using voltage and current values have been applied to locate the voltage sag source in a transmission network. In general the performance of the methods is much more accurate for symmetrical faults, where accurate results were obtained in 87% of the cases. On the other hand, for asymmetrical faults accurate results were obtained in 65% of the cases.

The analysis for meshed grids needs a broader idea of system boundaries. Hence, the concepts of upstream and downstream cannot be applied. More investigation is needed to develop a method to locate the sag source as inside or outside the system boundaries using wide area measurements techniques.

VII. ACKNOWLEDGMENTS

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IX. BIOGRAPHIES

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His work has comprised theoretical investigations at academic level, as well as extensive field measurements in power systems. Through the years he has been active in several Cigré and IEEE working groups. Dr. Karlsson is a member of Cigré and a senior member of IEEE. He serves as a reader at Chalmers University of Technology and he has supervised a number of diploma-workers and Ph.D students at Swedish universities.

Jaap Daalder received his DSc degree in power engineering from The Eindhoven University of Technology, The Netherlands. He was Associate Professor at the same university until 1984, when he left for Norway to become a Director of Technology and a Member of the Board of a subsidiary of the ABB Company in Skien. In 1993 he was appointed Full Professor at Chalmers University of Technology, Gothenburg, Sweden.

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